1. Introduction

- Project

  This project focuses on the analysis and performance evaluation of the RF front-end conversion receiver. The RF front-end plays a fundamental role in modern communication systems by receiving high-frequency radio signals and converting them into an intermediate frequency (IF), enabling efficient signal processing in the subsequent receiver stages. The implemented receiver architecture consists of several essential building blocks, including a Low Noise Amplifier (LNA), RF and IF filters, a mixer, and a Local Oscillator (LO), each of which has a significant impact on the overall system performance.

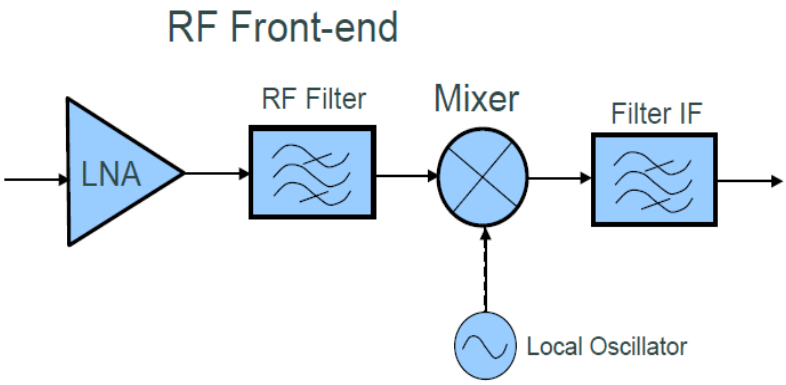
  The primary objective of this work is to investigate the performance of these RF front-end components through circuit-level simulations. Key performance metrics, including gain, noise figure, linearity, and receiver sensitivity, are extracted and analyzed to assess the contribution of each block to the overall receiver performance. This analysis provides valuable insight into the design of trade-offs and performance limitations of RF front-end architectures.

- Configuration of RF Front-End

  The RF front-end was configured according to the design specifications summarized below. These parameters define the operating conditions of the receiver and serve as the basis for all subsequent circuit-level simulations and performance evaluations.

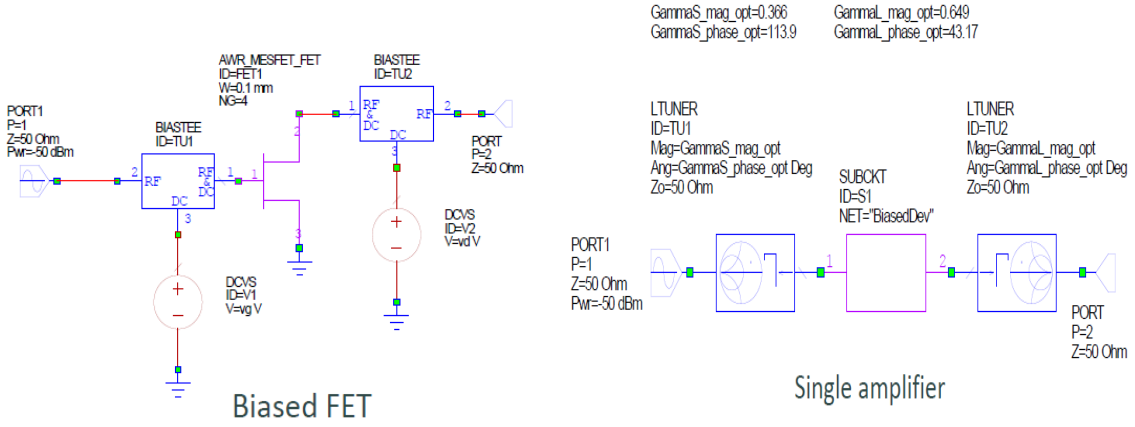
  - RF Carrier Frequency: 4.25 GHz
  - Modulation: 16-QAM with a data rate of 94.8 Mbit/s (Bandwidth of 32 MHz)
  - Intermediate Frequency: 500 MHz
  - Local Oscillator Frequency: 3.75 GHz

  Each RF building block—including the Low Noise Amplifier (LNA), Mixer, Local Oscillator (LO), and filtering stages—was characterized using key performance metrics such as gain, noise figure, and linearity. The combined performance of these components determines the overall behavior and efficiency of the receiver front-end.

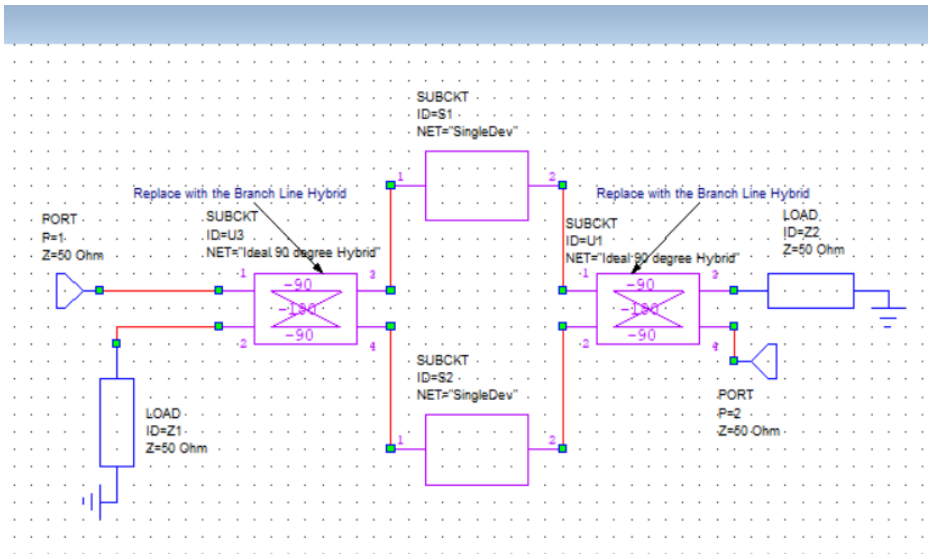


2. LNA

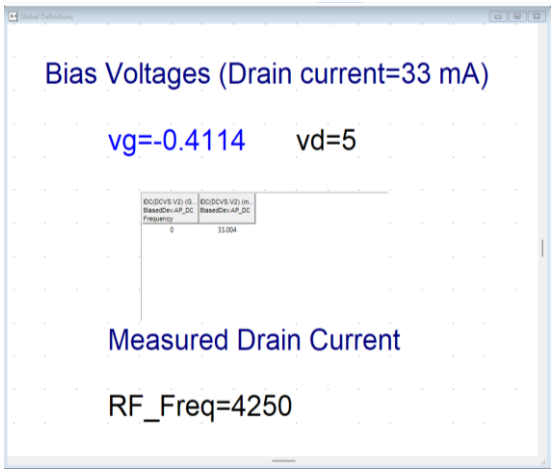
The LNA schematic is provided in the file "LNA_circuit.emp." The amplifier was implemented using a balanced architecture consisting of two identical FET-based amplifier stages modeled with nonlinear device characteristics in the Microwave Office (MWO). Each transistor was properly biased to ensure the desired operating point, while ideal impedance tuning elements (LTuner) were employed to achieve the optimum source and load reflection coefficients. This design approach provides proper impedance matching and enables the LNA to satisfy the required RF front-end specifications.



A balanced topology was adopted to improve the amplifier's overall performance in terms of gain, impedance matching, and linearity. The circuit employs two identical FET amplifier branches operating under identical bias conditions. The source and load matching networks were optimized by adjusting the corresponding reflection coefficients using ideal tuning components. This configuration ensures stable operation while maintaining efficient power transfer and high linearity across the operating frequency band.



Simulation Setup: To establish the desired operating point, the gate bias voltage was adjusted until a drain current of 33 mA was achieved, as illustrated in the figure below. Once the bias condition was established, a series of circuit-level simulations were performed to evaluate the LNA performance. The analysis focused on the extraction of key performance metrics, including transducer gain, noise figure (NF), the 1-dB compression point (P1dB), and the third-order intercept point (IP3). These parameters provide a comprehensive assessment of the amplifier's gain, noise performance, and linearity, allowing its suitability for the RF front-end to be evaluated.



To obtain a more practical circuit implementation, the ideal hybrid couplers were replaced with branch-line hybrids. The electrical characteristics of the branch-line hybrid were determined according to the design equations presented below. The hybrid was designed with a coupling coefficient of 0.5 and a characteristic impedance of 50 Ω, while its electrical length was selected as one-quarter of the operating wavelength. The calculated transmission-line dimensions were

subsequently incorporated into the LNA design to replace the ideal hybrid components. The characteristics of the branch line hybrid can be described using the following equations:

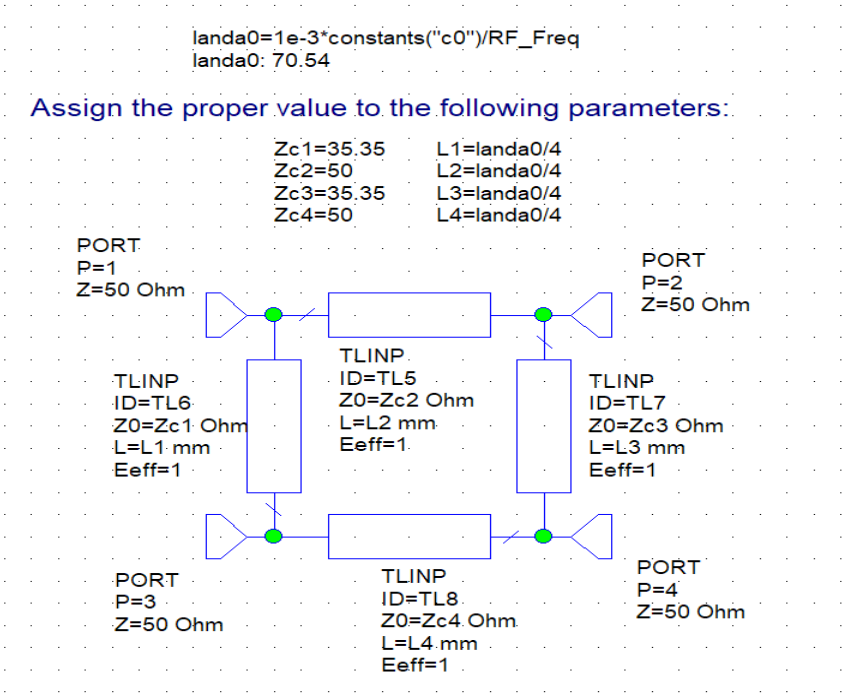
- The input impedance **Y'c** is given

$$Y_c' = Y_0 \frac{1}{\sqrt{1-C}}$$

- The output impedance **Y''c** is

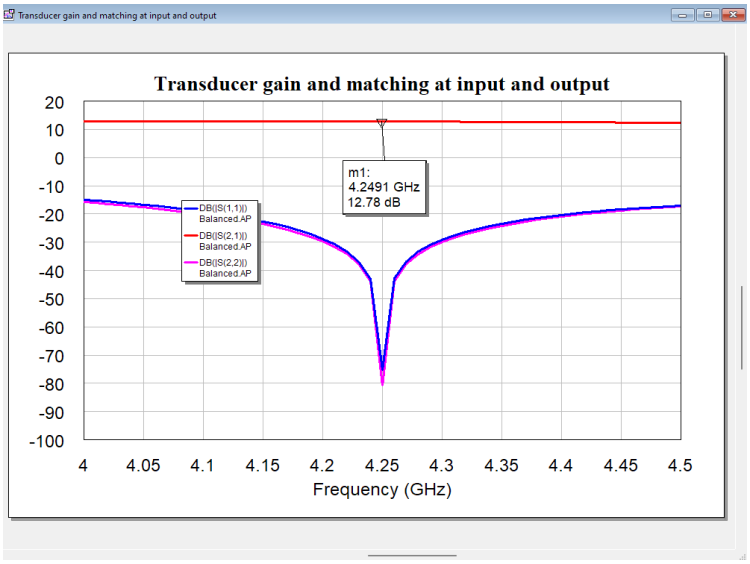
$$Y_c'' = Y_0 \sqrt{\frac{C}{1-C}}$$

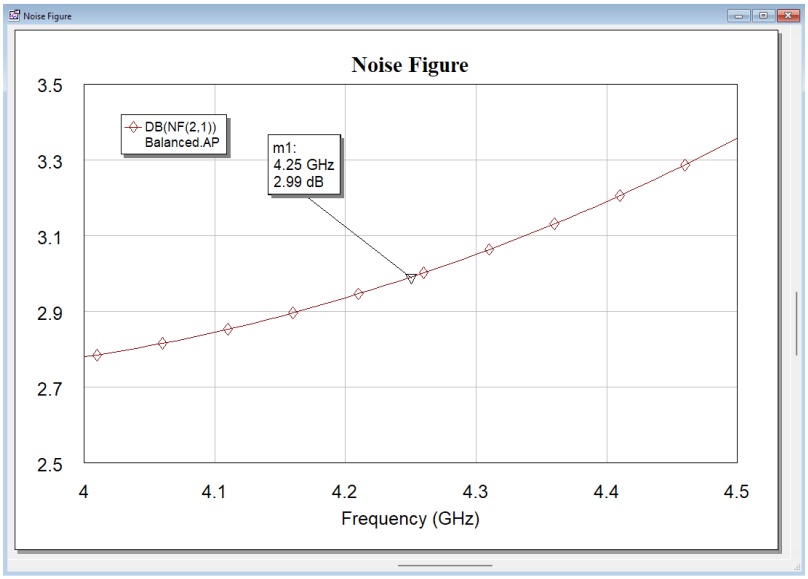
For the branch line hybrid, the coupling coefficient C is set to 0.5, and the electrical length is calculated to be $\frac{\pi}{2}$, which corresponds to one-quarter of the wavelength (approximately 17.635 mm). Using a characteristic impedance of Z0 = 50 Ω, we can now determine the values for the branch line hybrid to replace the ideal hybrid in the system.

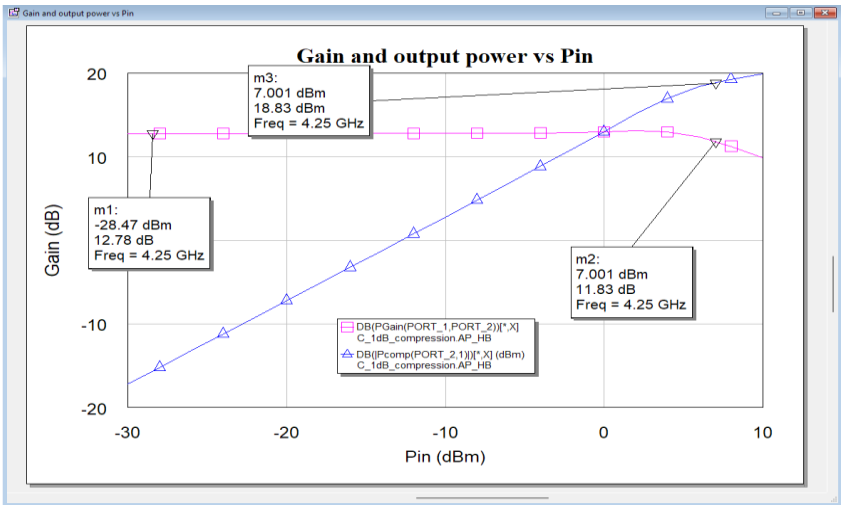


Circuit Evaluation of LNA Parameters:

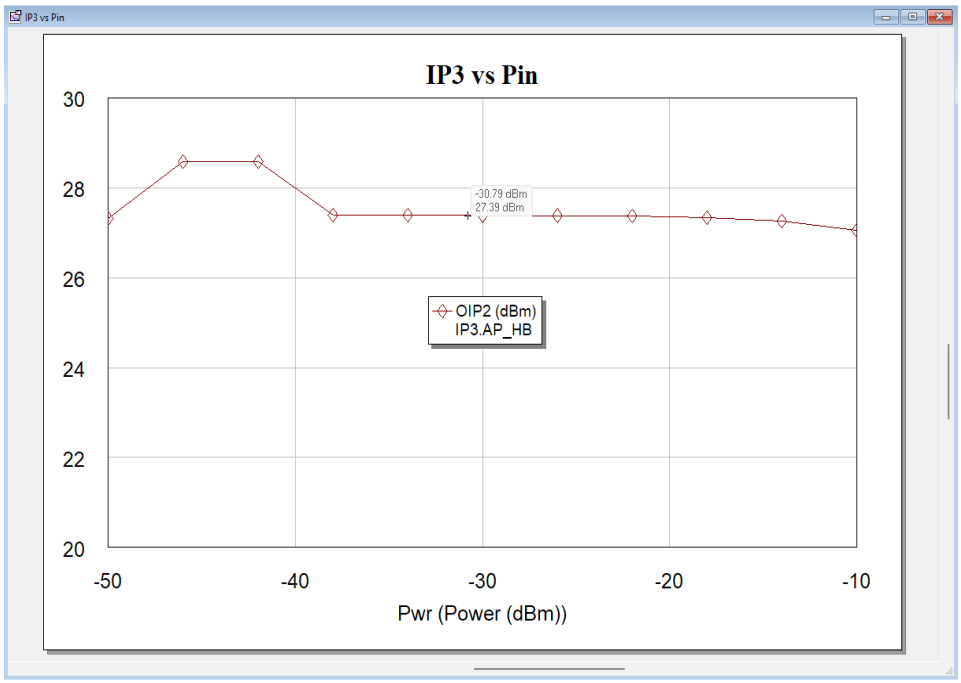
Following the circuit-level simulations, the LNA performance was evaluated by extracting the relevant parameters from the simulated characteristics. The transducer gain was determined from the Transducer Gain and Input/Output Matching response, which was simultaneously used to verify the input and output impedance matching at the operating frequency. As shown in the figure below, the LNA achieved a transducer gain of 12.78 dB at the design frequency of 4.25 GHz.







According to observing the figure, the IP3 value is equal to 27.39 dB.



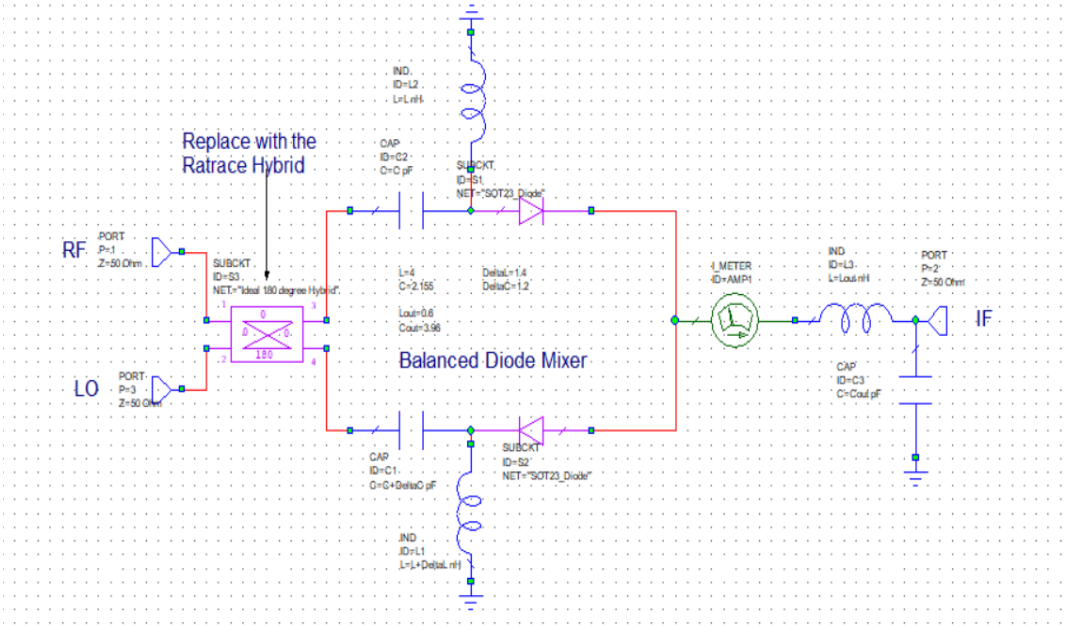
Circuit Evaluation of LNA Parameters:

- Gain (LNA): 12.78 dB
- Noise Figure (NF_LNA): 2.99 dB
- Power at 1 dB Compression (P1dB_LNA): 18.83 dBm
- Third-Order Intercept Point (IP3_LNA): 27.39 dBm

3. Mixer:

The mixer schematic is provided in the file "Mixer_circuit.emp." The mixer was implemented using a balanced diode topology driven by a 180° hybrid coupler. To obtain a practical implementation suitable for microwave realization, the ideal hybrid was replaced with a Rat-Race hybrid, whose dimensions were determined according to the design specifications. All remaining circuit components were retained without modification to ensure a consistent comparison between the ideal and practical implementations.

Although not physically integrated into the mixer circuit, an Intermediate Frequency (IF) filter was included in the simulation environment to suppress unwanted mixing products and isolate the desired IF component. This configuration enables an accurate evaluation of the mixer's frequency conversion characteristics and its contribution to the overall RF front-end performance.

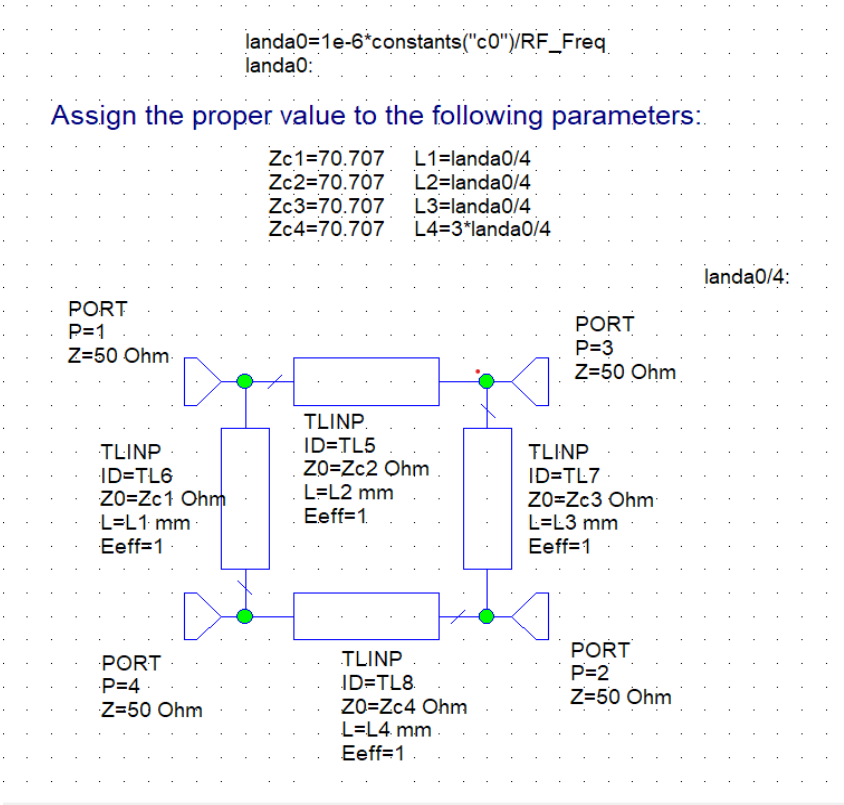


Balanced Configuration with Rat-Race Hybrid:

A balanced mixer configuration was adopted to improve signal isolation and suppress undesired signal components. The initial design employed an ideal 180° hybrid coupler, which was subsequently replaced with a practical Rat-Race hybrid to achieve a realizable microwave implementation. Compared with the ideal hybrid, the Rat-Race coupler provides improved port isolation while maintaining the required 180° phase difference for balanced mixing operation.

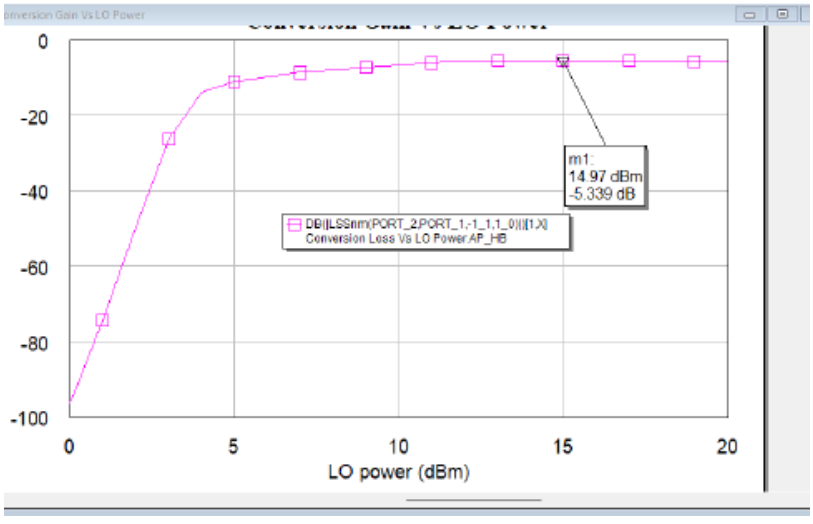
The Rat-Race hybrid was designed for a characteristic impedance of 50 Ω. Accordingly, the impedance of each transmission-line section was calculated as 70.707 Ω. The lower section of the hybrid was designed with an electrical length of 3λ/4, whereas the remaining sections were implemented with an electrical length of λ/4, following the standard Rat-Race hybrid design

methodology. The calculated dimensions were then incorporated into the mixer circuit for subsequent performance evaluation.

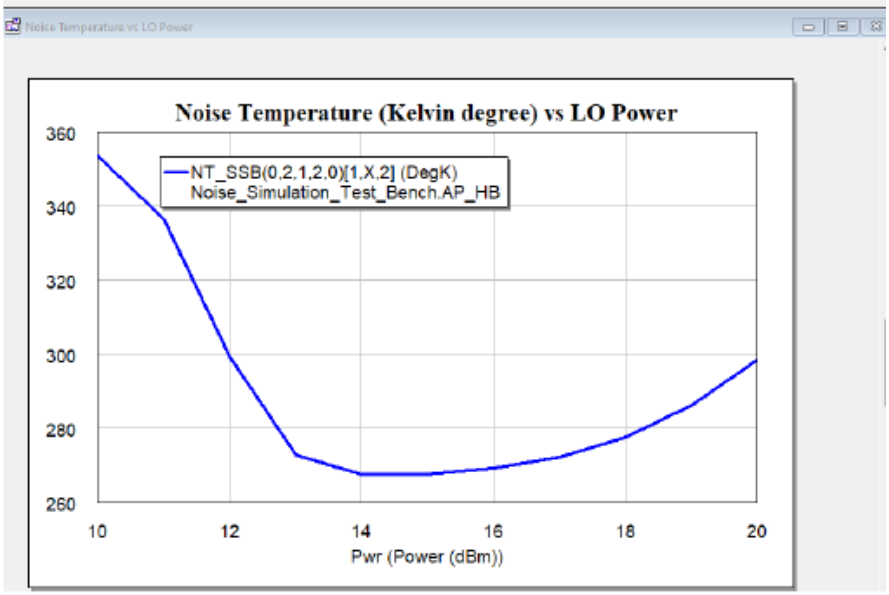


Circuit Evaluation of Mixer Parameters:

According to the graph we can see the value of Conversion Gain is equal to -5.339dB



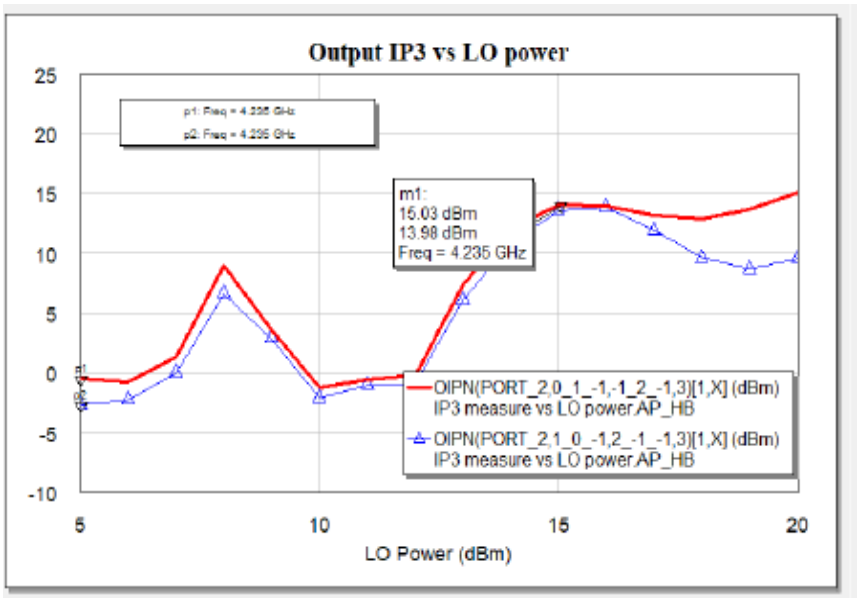
This value of TSSB based on the graph is equal to 267.7 kelvin degree



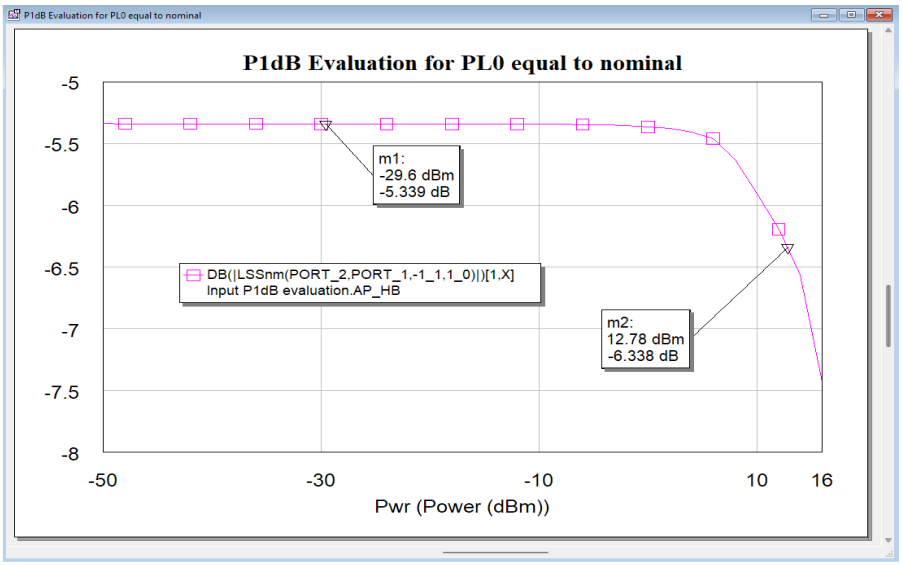
According to the graph below, the value of IP3 is 13.98.

□□□□ = □□□ − □□□□□□□□□□ □□□□

13.98dBm − (−5.339dB) = □□. □□□□□□



By entering the value of LO POWER inside the simulator, the value P1dB =12.78 dBm is read.



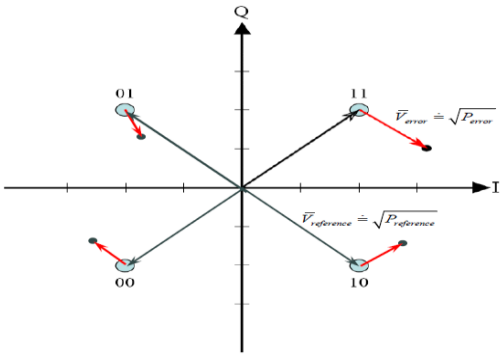
Circuit Evaluation of Mixer Parameters:

- Conversion Gain: -5.34 dB
- TSSB (Single Side Band Noise Temperature): 267.8 K
- Input IP3: 19.31 dBm (calculated from Input IP3 minus Conversion Gain)
- P1dB: 12.78 dBm

Error Vector Magnitude (EVM) Evaluation:

The Error Vector Magnitude (EVM) was evaluated to determine the acceptable dynamic range of the RF input power while maintaining the required signal quality. EVM represents the deviation between the ideal constellation points and the received symbols, making it a key metric for assessing the overall performance of digital communication systems.

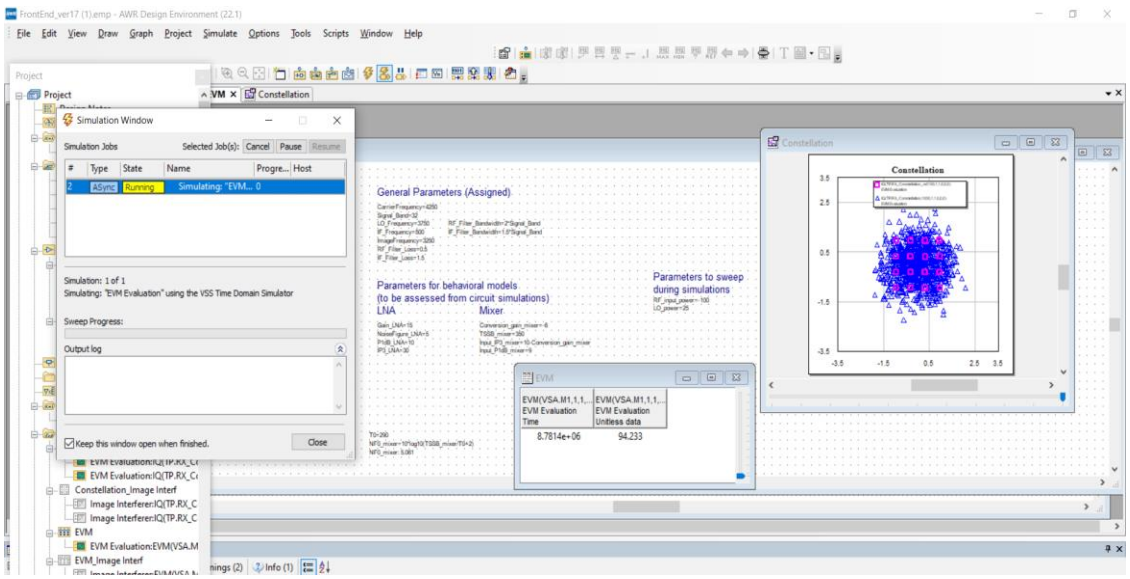
The RF input power was varied over a wide range, and the corresponding EVM values were obtained through circuit-level simulations. The results indicate that the receiver maintains an EVM below 5% for input power levels ranging from −73.2 dBm to 0 dBm. Within this operating range, the RF front-end exhibits linear behavior with negligible distortion, ensuring reliable signal demodulation under ideal operating conditions.

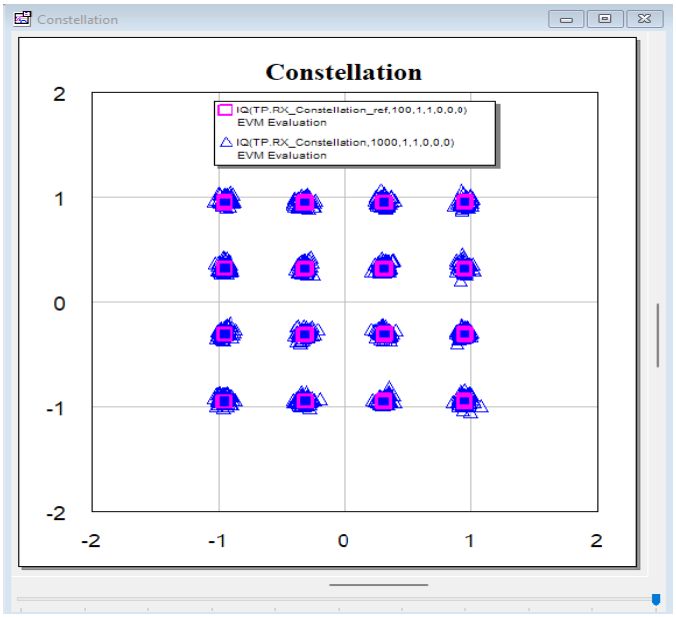


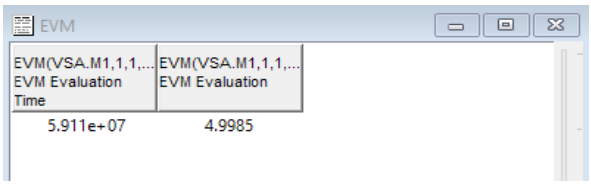
Constellation of
4-QAM modulation

$$EVM(\%) = 100 \cdot \sqrt{\frac{P_{error}}{P_{reference}}}$$

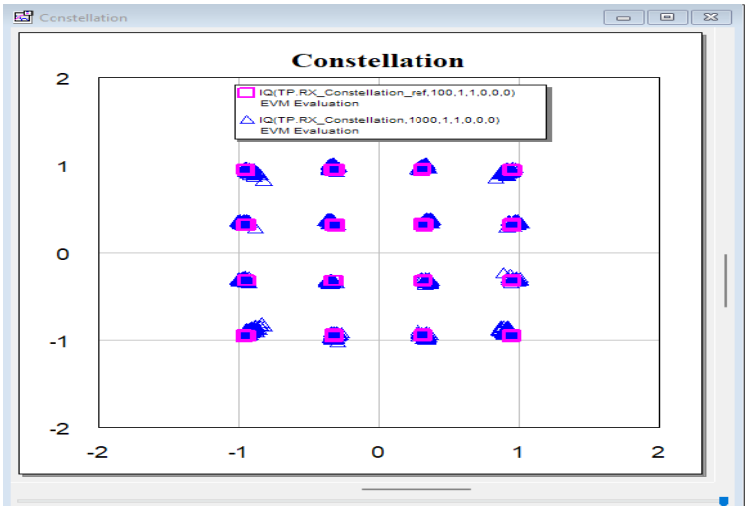
$$EVM(dB) = 20\log_{10}\sqrt{\frac{P_{error}}{P_{reference}}}$$

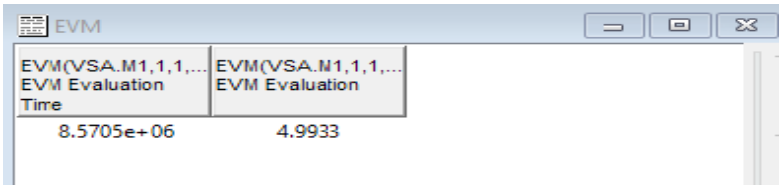


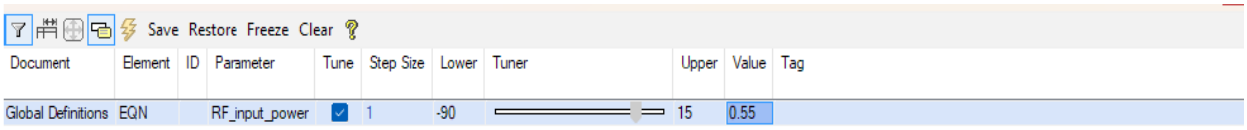




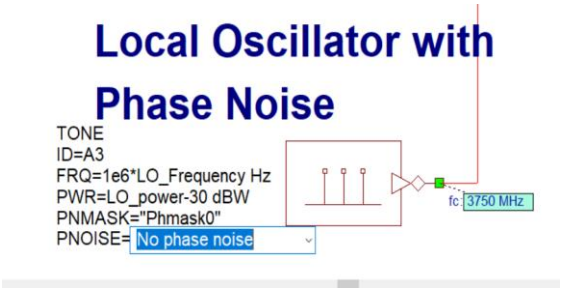
Afterwards, we increase the input power until we obtain a very similar EVM value.

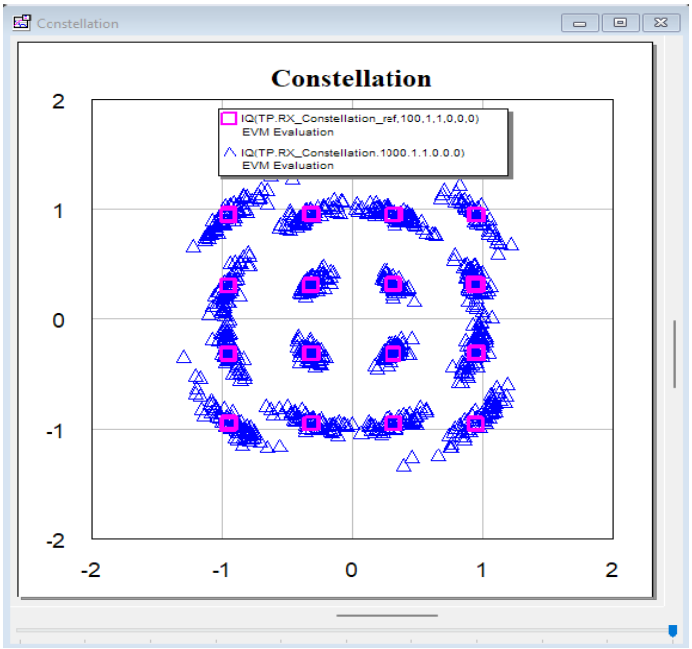


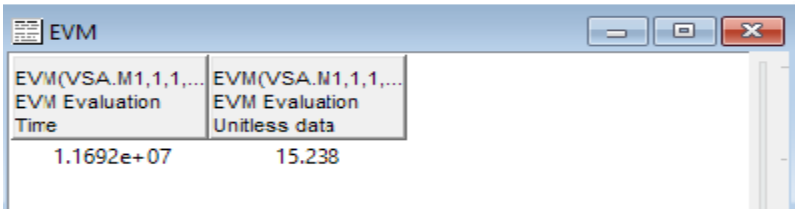


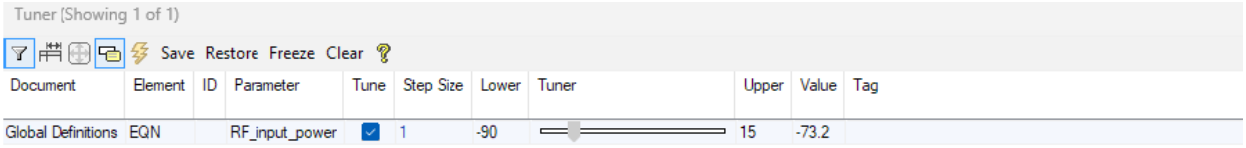


The constellation diagram exhibits a slight deviation from the ideal symbol locations as the RF input power increases. Although the received symbols remain clearly distinguishable due to the relatively high signal power, their accuracy is degraded by the nonlinear behavior of the RF front-end. Up to this stage of the analysis, the effect of local oscillator (LO) phase noise had not been considered. To investigate its impact, phase noise was subsequently introduced into the LO model by applying the first phase-noise mask. The resulting degradation in the constellation diagram provides a clear indication of the influence of oscillator phase noise on the received signal quality.

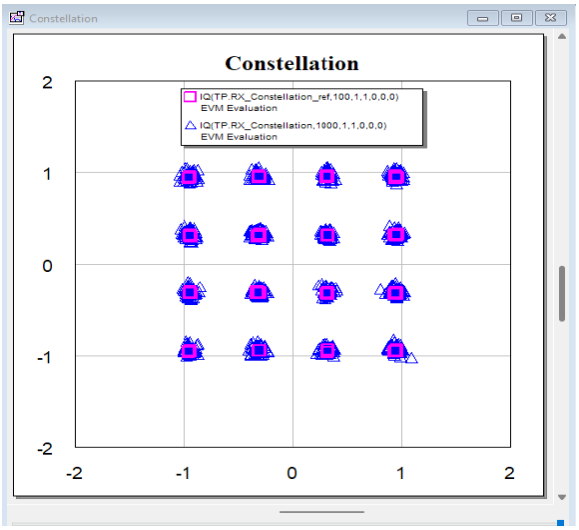


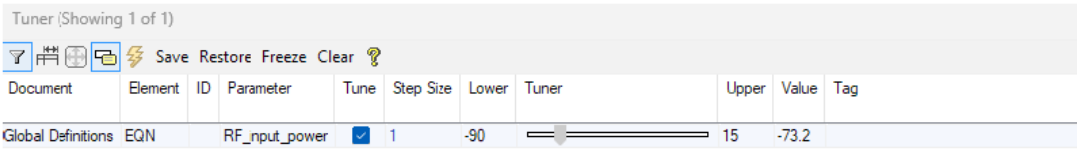






Phase noise causes rotation in the received symbols and increasing the power cannot mitigate this issue. As a result, the EVM increases. To address this, we need to select a phase mask that minimally affects the EVM. After simulating different phase masks, the third mask was found to be the most suitable option.





To evaluate the robustness of the receiver against image-frequency interference, a jamming signal was introduced at the image frequency of the mixer. The objective was to determine the maximum interference power that could be tolerated while maintaining the EVM below 5.2%. The power level of the image interferer gradually increased until the EVM exceeded the specified limit. The simulation results indicate that the maximum acceptable image interference power is approximately −40 dBm, beyond which the receiver performance begins to degrade significantly.

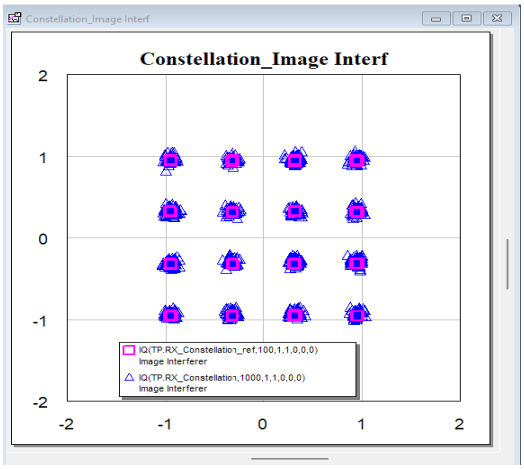
To investigate the influence of oscillator instability under practical operating conditions, phase noise was introduced into the Local Oscillator (LO) model using different phase-noise masks. The resulting EVM was evaluated for each case to quantify the impact of phase noise on the received signal quality. Among the evaluated masks, Phmask3 provided the best overall performance, maintaining the EVM within the acceptable limit of 5%. These results indicate that the selected phase-noise profile introduces only a limited degradation in signal quality and is therefore suitable for the designed RF front-end.

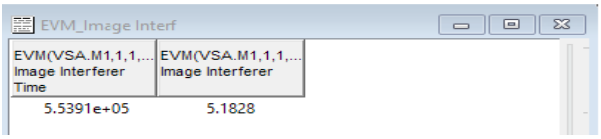
Image interference arises when an undesired signal is present at the image frequency of the mixer, producing an additional frequency component at the intermediate frequency (IF) after the mixing process. Ideally, this unwanted component is suppressed by the RF image-rejection filter; however, the effectiveness of this suppression depends on both the filter selectivity and the relative power levels of the desired and interfering signals.

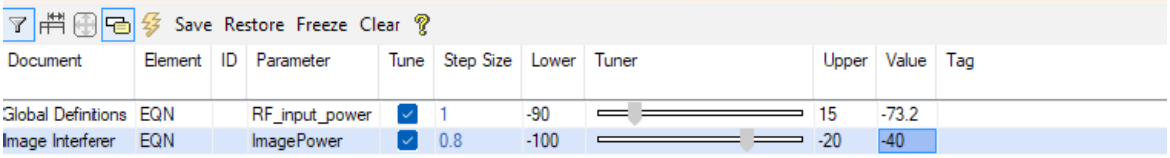
Image Interferer Evaluation:

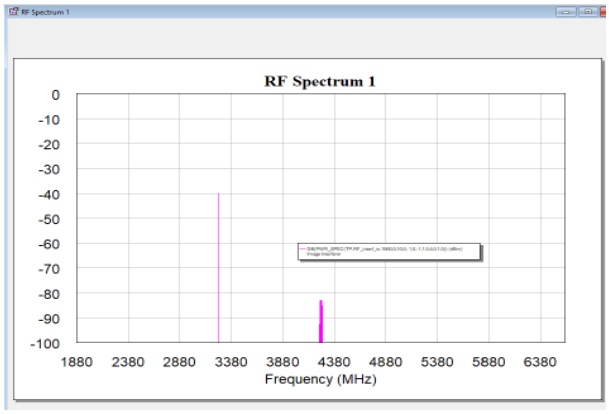
To quantify the receiver's immunity to image-frequency interference, the power level of the image interferer gradually increased while monitoring the resulting Error Vector Magnitude (EVM). The objective was to identify the maximum interference power that could be tolerated without violating the specified EVM limit of 5.2%.

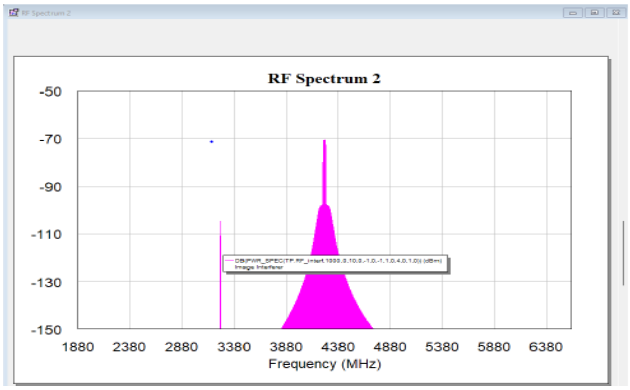
The simulation results indicate that the receiver can tolerate an image interferer power of up to −49 dBm. Beyond this threshold, the EVM exceeds the acceptable limit, indicating a noticeable degradation in signal quality.











Results:

Parameters for behavioral models
(to be assessed from circuit simulations)

LNA	Mixer
Gain_LNA=12.78	Conversion_gain_mixer=-5.34
NoiseFigure_LNA=2.99	TSSB_mixer=267.8
P1dB_LNA=18.83	Input_IP3_mixer=13.97-Conversion_gain_mixer
IP3_LNA=27.4	Input_P1dB_mixer=12.78

Parameters to sweep during simulations

RF_input_power=-73.2

LO_power=15